

The Manasseh Hill Country Survey

Volume II: The Eastern Valleys and
the Fringes of the Desert

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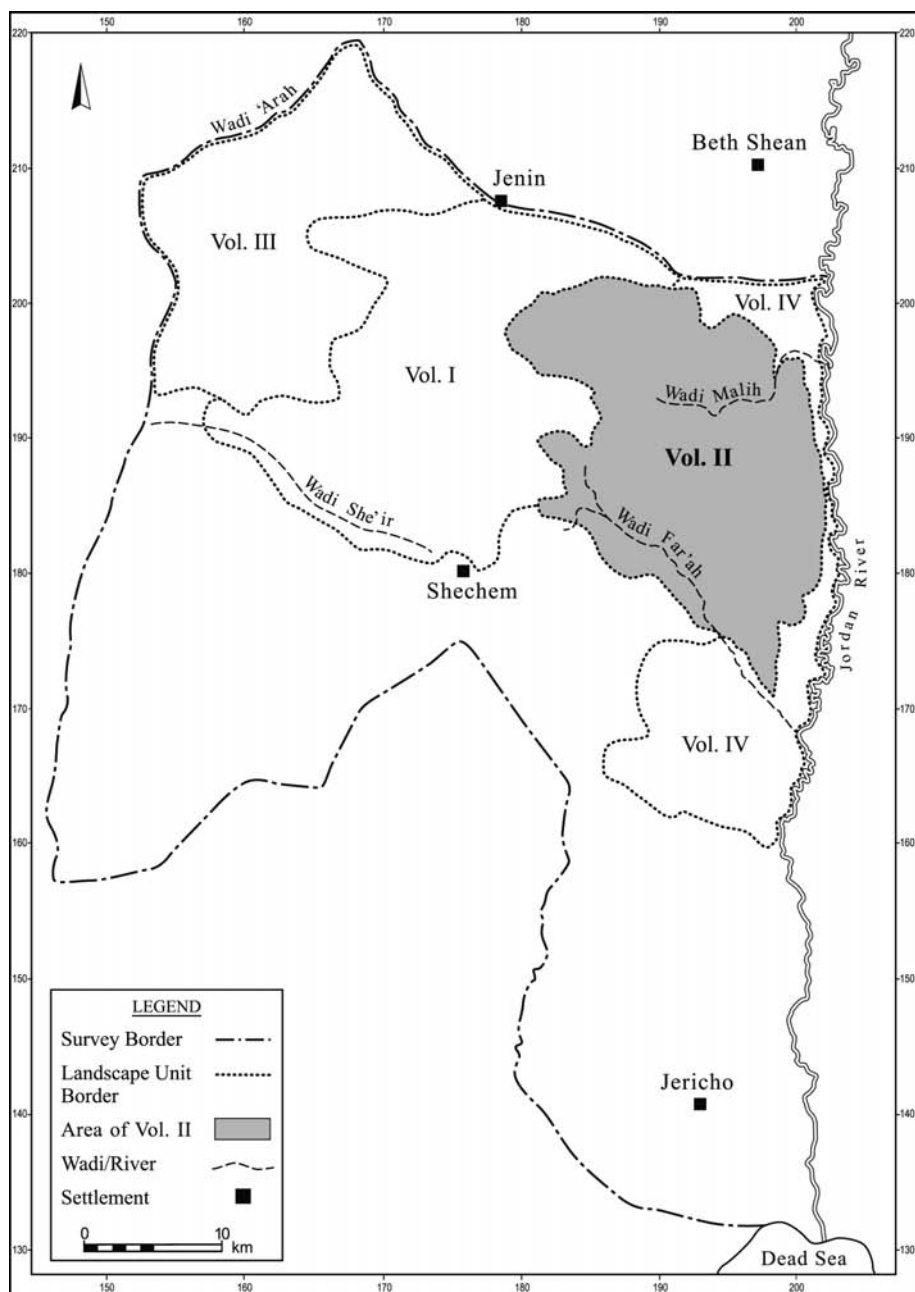
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The Area of the Manasseh Survey, divided into volumes of publication.

B. The Geography of East Manasseh

1. *Boundaries of the Region*

This area, encompassing almost 540 sq km, has clear, natural borders in the south and east; in the north and west we relied on the borders of the relief lines (geomorphologic lines) which require an explanation.

The northern border: This border passes along a tributary of Wadi Shubash, beginning its course in Zebabdeh Valley, south of the village of Tilfit. The border between the territory surveyed in the first volume (the Shechem Syncline) and the second volume passes, at its inception, along the road from Zebabdeh to the village of el-Mughair. One km to the east of the place at which the paved road to Kh. Tannin leaves the above road, the border returns to the course of Wadi Shubash. In the east the northern border ends at the place where this wadi exits to the plain of the Beth Shean Valley (Israel Grid 2002/1920), not far from the Qa'un Springs.

The eastern border: This border passes along the line where the mountainous ridges of the Desert Fringes meet the narrow plain of the Jordan Valley. In most places the border passes along the Beth Shean – Jericho road, but an exception is the northern portion of the survey's territory, for about 4 km,

where an 'enclave' near the village Bardaleh entered the area of the Fourth volume. Some of the wadis to the west of the Beth Shean – Jericho road have been left out of the present volume and were included, also, in the next one.

The southern border: This border is based on the most important formation of the area, the geological fault of Wadi Far'ah. The beginning of the borderline, in the east, is at the Jordan River itself, near Jiftliq Junction. From there it passes, along the contour line between Ras el-Kharubeh and the river delta, to the narrow crossing point of Bab en-Naqb. From there the borderline crosses the river southward and crosses the slopes of Jebel Kebir, up to the summit of Sheikh Bilan near Elon Moreh. From there the border turns north and ends at 'Ain Beidan.

The western border: This border is based on the inner, eastern margins of the Shechem Syncline. The line goes along the road from 'Ain Beidan to ez-Zebabdeh (our road B5). This landscape unit includes the villages of Sir, Kfer and ez-Zebabdeh itself, and the border passes along the eastern margins of Sanur Valley. From there it ascends through Kh. 'Anahum and joins up with the northern border near Tilfit.

2. *Geology and Relief*

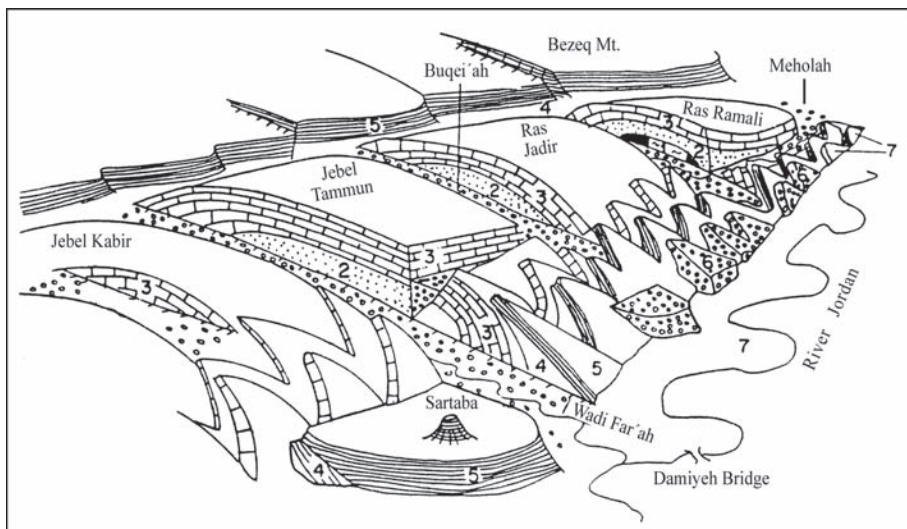
General: The main geomorphologic formation of the area is the Far'ah Anticline, which borders the Shechem Syncline in the east (Cook, Roth and Meimaran 1970; 1980). The anticline is the northern continuation of the chain of anticlines of Judah and Ephraim (Picard and Golani 1965). It contains a broad, subterranean axis, with a relatively narrow upper axis. The slanted blocs of the Far'ah Anticline slope moderately west, while the eastern branch drops sharply toward the Jordan Valley.

The length of the anticline is 30 km, and its width reaches about 15 km. It includes Jebel Kebir, Jebel Tammun, the northern Buqei'ah, Ras Jadir and Ras el-Mubarah.

With the exception of the Buqei'ah, all the blocs slope with their summits facing southwest.

The anticline is broken by faults which created river valleys and wide ravines. The longest of the faults is the valley of Wadi Far'ah. To the north, along the southeast–northwest axis, stretch the Buqei'ah and the valley of Wadi Malih. In all these areas, the soft, unstable rock formations aided the erosive processes.

The special structure created a relatively isolated geomorphologic units in East Manasseh. The absolute elevation of the blocks of the Far'ah Anticline is just 500-800 m a.s.l., but the breaking and transferring processes caused their considerable elevation and strong erosion



2. A tri-dimensional geological section of East Manasseh. The figures are the following groups: 1 – Arad; 2 – Kurnub; 3 – Judah; 4 – Mount Scopus; 5 – Avdat; 6 – Tiberias; 7 – Dead Sea (after Meimaran 1992, 365).

eastward, to the base of erosion of the Jordan Valley. These processes greatly affected the creation of the Desert Fringes.

The fringes of Manasseh, the plateau on the eastern edges of the anticline, form a triangular shape whose southern point is at el-Mahruq near Jiftliq, whose northern point is at the place where Wadi Shubash flows into the Jordan and whose eastern point is north of Sahl Ibziq. This plateau contains four large *grabens* or depressions, which point toward the Jordan Valley. These are the result of faults and folds which fractured the anticline and were later widened by erosion. From north to south are the following recesses: Wadi Shubash, Wadi Malih, the Buqe'ah and Wadi

Far'ah.

3. *The Geology of East Manasseh* (summary of the article by Y. Meimaran 1992)

In East Manasseh an almost entire series of rock units has been exposed, from the Upper Jurassic Period to the present time. The rocks are divided into formations which are divisible into groups. The overall depth of the seven rock groups is more than 2.5 km, and these are the main groups:

Arad group: This group of the Jurassic Period is composed of limestone rocks with a few intermediate layers of Huwwar. Remains of shells and sea urchins indicate shallow sea conditions. The group, 200 m



3. A view to the west, to the Wadi Malih cliffs. The Chalcolithic site of 'Ain Hilu is on the left slope. 1990.

deep, is exposed near Burj el-Malih. The layers are concentrated here in an elevated bloc.

Kurnub group: This group, dated to the lower Cretaceous Period, is composed of sandstone with remains of wood, and chalky sandstone with remains of shells. The wood and sandstone indicate stratification in land conditions, following sea withdrawal at the end of the Jurassic Period. The chalky sandstone is the result of the next step: withdrawal of the sea, with the stratification of the Judah group. The Kurnub group, exposed close to the Arad group near Burj el-Malih, is 200 m deep. Its layers were also detected to the south, at the edges of Jebel Tammun and Jebel Kebir.

Volcanic rocks: The base of the Kurnub group is composed of the special "Taiyasir Volcanics", made of magmatic rocks.

Judah group: Dating to lower Cretaceous-Turonian periods, this group is 1000 m deep and built of carbonate rocks. These consist above all of limestone and Dolomites, with few Huwwars and chalks. These were stratified in a shallow sea. At the base of the group there is a "blanche" limestone cliff, identified in the ranges of Ras Mubarah, Ras Jadir and Jebel Tammun and Jebel Kebir.

Mount Scopus group: This chalky group is dated to the Sennonian-

Paleocene Periods. The microscopic fossils in the chalks indicate a sea hundreds of meters deep. The upper side of the group consists of Taqiyeh formation, whereas its middle is composed of flint layers some centimeters to few meters deep (Meshash formation). The average depth of the group is 300 m, with points where it is only few meters.

The exposures of the group are concentrated along the "seams" between the anticlines and the synclines. These, for instance, are found between the Far'ah Anticline and the Shechem Syncline.

Avdat group: This is dated to the Eocene Period. It is built of limestone and chalk with some quartzite, 350 m deep in East Manasseh to 600 m deep in the Gilboa. Its rocks contain microscopic fossils originating in a shallow sea.

Tiberias and Dead Sea groups: These were stratified in the Neocene-Quaternary Period, when the processes of breaking and folding began. These young groups were stratified in the lower areas of Manasseh. The Tiberias group is based upon a rough conglomerate of pebbles that originated in the Avdat-Kurnub groups, which moved through erosion. On top of it there are sandstone layers replaced by limestone, chalk and gypsum. The overall depth of the Tiberias group is 150 m and it extends along the Desert Fringes.

4. *Climate and Soils*

Of the four climatic types discussed here, three – Mediterranean fringes, semi-arid and tropical (Sudanese-Decanese) – are typical only to East Manasseh. The fourth (Mediterranean) appears in the eastern valleys.

In the eastern valleys and the Far'ah Anticline the climate is Mediterranean (*cf.* Zertal 2004, 26), gradually becoming arid toward the east. East Manasseh is a typical transitional area between the Mediterranean and the arid climates. East of the national watershed, precipitation drops in the direction of the Jordan Valley (Goldreich 1977, 31 and fig. 2). The 600-mm isohete

(average rainfall) on the watershed falls to 300-mm in the Jordan Valley. This drop is gradual, due to the movement eastwards of the rain clouds through the passes of the Far'ah ranges (Goldreich 1977, 30 and fig. 1). The overall result is an average rainfall of 350 mm in East Manasseh.

Droughts are less frequent here than in the fringes of Ephraim and Judah. Like other arid regions, East Manasseh shows a high correlation between climatic and agricultural drought. The rain here is unequally divided over seasons and years (Halpern 1966, 53 *ff.*). This phenomenon finds expression in the Buqei'ah, where unequal rains lead to sowing every other year (oral



4. An air view of Kfer, a small village in the Zebabdeh Valley. Around the place the alluvial plains of the valley are seen. 1993.

communication, Ali Abdallah of Tammun).

The summer in East Manasseh is long and hot, with high temperatures and strong winds along the wadi-valleys. The Sudano-Decanese climate in Wadi Far'ah and (to an extent) Wadi Malih is typified by high temperatures in summer, many hot days and low humidity. The winter is moderate, with low temperatures and few rains. There is also unequal division of rains between the months and the years (Halpern 1966, 51-61).

According to Thorntwhite (Goldreich 1977, 29-33), East Manasseh, Wadi Far'ah and the Jordan Valley all belong to arid region E.

Climatic conditions make agricultural cultivation difficult. Wheat and barley are sown only during rainy years, but even then, no more than 10%-20% of the area is cultivated. This raises the problem of agriculture in periods of prosperity in East Manasseh – MBA II, Iron Age II and Byzantine Period. Could agriculture have been the economic basis of these societies?

The different soils of Manasseh have been reviewed thoroughly in the first volume of the survey (Zertal 2004, 26-27). Here we concentrate upon complementary details relevant to the soils of East Manasseh.

Terra Rossa appears mainly on

the Far'ah range - the upper plateaus of Ras Jadir and Ras es-Salmeh. However, the terra rossa areas here are not consecutive and appear in "patches". This may evolve from the low quantity of precipitations which is involved in the creation of that kind of soil. Terra rossa, together with brown grumosoles, is found also in parts of Wadi Far'ah, the Buqe'ah and the hills around it.

Rendzina soils extend over large parts of the Desert Fringes (Dan 1977, 15): above all in the LU's of Wadi Malih, Ras Jadir and the Desert Fringes. Similar areas are situated near the outlet of Wadi Shubash to the Beth Shean Valley. There, eroded chalky areas are the basis to the rendzina soils. These are dark and light rendzina, which develop upon Nari rocks and are not very fertile.

Brown Mediterranean forest soil is highly typical to East Manasseh. Most of the wide areas from Jebel Tammun to the Buqe'ah and to the Jordan Valley are covered by this soil. Sometimes it is combined with rendzinas. Ravikovitz (1981, 76) claims that the brown soil is created above all in humid climate. Its existence in the arid areas of East Manasseh may indicate extreme rainy conditions in the past.

Eroded alluvial soils (grumosoles) are located in the eastern valleys, the Buqe'ah and the

northwestern part of Wadi Far'ah. These soils are deep, dark and with A-C section. The soils are relatively fertile, but low air capacity makes them suitable mainly for growing cereals. In the valleys of Tubas and Zebabdeh there is a difference between their western and eastern edges. Olives are presently grown in the former, when cereals are sown in the latter. The limit for agriculture in East Manasseh, therefore, is climate and not soils. This, most probably, was the situation in the past, as well.

Colluvial-alluvial soils extend in the wadis and the margins of hills, neighboring the rendzinas and terra rossa areas. Most of them are deep

and dry, but their fertility depends upon their origins and the percentage of organic material.

Stony desert soils (desert lithosoles) are found in the regions close to the Jordan Valley, which typify the eastern parts of East Manasseh and the deltas of the wadis Malih, Far'ah, Shubash and the other small wadis which open to the Jordan Valley. These soils occur in patches and have a high percentage of rocks and stones.

5. Water and Vegetation

The water sources have been generally reviewed in the first volume (Zertal 2004, 27-31); only those in East Manasseh ("the eastern water



5. The delta of Wadi Far'ah, looking south. The dark areas are irrigated soils, a typical phenomenon of the wadi along many periods. In the background the Sartaba hills are seen.

strip”) are discussed here.

The eastern strip contains of the following springs:

1. The Ballatah Springs, no. 100 in the list (*cf.* Zertal 2004, 579-581);

2. ‘Ain Kekub (no. 82 in the general list);

3. ‘Ain Beidan (no. 83 in the list);

4-5. The springs of Dilb and Far‘ah, nos. 84-85 in the list.

The aquifers and aquicluds (water-bearing layers) of the eastern strip are similar to those in the west (Zertal 2004, 28), so that springs evolve from the same geological features. But unlike the western strip, these springs do not continue to northeast, along the “seam” between the Eocene and Sennonian areas. Northeast of ‘Ain Far‘ah there are no more springs, due to the inclination of the geological strata. Under theoretical conditions of more precipitations, more springs along the “seam” can be proposed. This model can probably explain concentrations of sites, above all during the Middle Bronze Age II period.

The few eastern springs are relatively strong, with a high water capacity. The springs of Beidan, Dilb and Far‘ah, all sources feeding Wadi Far‘ah, are among the most powerful in Manasseh. Two other groups of springs are situated near Wadi Far‘ah and Wadi Malih. Near the

springs south of Wadi Far‘ah (Juwar, Dabbur, Za‘anuni, Miski and others) many sites developed, in spite of their low water capacity. In the lower part of Wadi Malih there are several springs (Meiyiteh, Malih and Hilu) near which large sites were developed.

The two following perennial wadis which cross East Manasseh had deep influence on the history of settlement:

Wadi Far‘ah (Nahal *Tirzah*) is the longer and more important one. It is 37 km long, from ‘Ain Far‘ah to the Jordan River near Jisr Damiyeh. Boneh and Baidah (1977, 46) state that its yearly capacity (of the springs Dilb and Far‘ah) was 8.5 millions cubic m. ‘Ain Beidan supplied 4 million more. The small springs south of the wadi do not add much more.

The various characteristics of Wadi Far‘ah – length, moderate slope, stability of stream and fertile soil – made it the richest and best irrigated part of Manasseh. Irrigation is based, today as in the past, upon channels and aqueducts. Information on irrigation here exists from the 18th century onwards. Guérin (original 1875; reprint 1969, 233-243) mentions an “ancient channel that leaves the river”. He describes, “... we crossed an ancient, man-made canal that took its waters from Wadi Far‘ah” (*idem*, 241).

When exactly such irrigation

began is hard to ascertain, but the continuity of settlement in Wadi Far'ah may indicate its long use. The wadi was also an important road, and see below.

Wadi Malih is much shorter than Wadi Far'ah and its water course is rather short. Its overall length is 16 km, of which 10 km bear water. It begins its stream near Burj el-Malih (Israel grid 1943/1923). The Meiyiteh Springs are sweet, but not far from it the salty waters of 'Ain Malih enter the wadi. Ilan (1973, 348) mentions a built aqueduct from 'Ain Meiyiteh to 'Ain Malih, which we were not able to find.

East of the Malih Springs the sweet Hilu Springs are located. Ir-

rigation here was possible in the small valley near Tel el-Hilu (no. 90). It seems that Wadi Malih was used above all to water the herds. Irrigation was negligible, due to the saline water and the small areas of land. Wadi Malih, like Far'ah, was also an important road.

The five vegetation groups of Manasseh, suggested by Zohari (1980, 186-190, and cf. Lipschitz 1987), extend over upon two phytogeographic regions: Mediterranean and Irano-Turannian (arid). Dannin (1970, 5) defines three long strips of vegetation in East Manasseh, with four groups. On the basis of the different scholars' opinions, East Manasseh is divided



6. *Pistacia Atlantica* tree east of Kh. Jabaris. These trees are a rare component in the vegetation of the Desert Fringes.

into the following sections:

A. **The Eastern Valleys** are part of the alluvial valleys' strip. In the first volume of the survey we have suggested (Zertal 2004, 30-31) – following Zohari – that the climax (original plant group) in the valleys was forest park of the Thabor oak. Following the cutting down of the forest to make room for cultivation, a secondary group, typified by *Prosopidion Farcta*, took over the valleys.

B. On the **Far'ah Anticline**, comprising Jebel Kebir, Jebel Tammun, Ras Jadir and Ras Mubarah, there are Mediterranean fringe groups, with forest of carob and Pistacia. These groups stretch on the transitional strips between the Mediterranean and the arid zones, 3 to 15 km wide. Here again, the climax group is the forest park of carob and Pistacia. Among the trees grows yearly vegetation, providing good grazing for herds.

C. East of the anticline (**the Desert Fringes**) is the realm of Irano-Turannian vegetation. Here no trees are found (Sabah 1992, 50), and the only "tree" (or high bush) is the *Sisyphus*. Dannin (1970, 127) claims: "This bush is supposed to be an intruding plant, that has penetrated after the destruction of the original vegetation. In East Manasseh, there is no other tree to be "replaced" by the *Sisyphus*. Therefore we may assume that this bush is original here."

The lack of trees is somehow strange. A yearly average of 350 mm rain can easily be the basis for tree growing, at least in the low areas and the wadis. In the Negev Highlands, for example, trees grow in areas more arid than ours. The complete lack of trees, therefore, needs further exploration.

Traditional agriculture is seasonal and very limited today. No orchards cultivated by the local Arabs were found. However, the new Jewish settlers in the same region (*cf.* the vineyards of Beqa'ot and Ro'i) successfully grow fruit orchards. The finds of our survey indicate olives and grapes which were grown in the past (Iron Age II-III, Byzantine and Early Moslem Periods). The overall picture, though, is of negligence and over-grazing of the region. This process seems to have begun during the Middle Ages.

D. **Wadi Far'ah** is typified by a perennial water supply and high temperatures – a tropical climate. The vegetation here is Sudano-Decanese in type (Zohari 1955, 254; 1980, 187). The groups are governed by the Egyptian *Zaquum*, the *Sisyphus* and the "apple of Sodom". They stretch from the lower parts of the wadis of Far'ah and Malih to the Jordan Valley. Under these conditions, cultivation and irrigation are possible, together with watering of flocks. Agriculture here seems to concentrate on palm orchards and supply green pasture all

year round.

6. *The Roads and Paths (Fig. 8)*

A. Roman and Byzantine Roads

During the survey several Roman roads were recorded; some were already known and others were newly found. "Roman road" is defined as a built road with rim stones. All of them are indicated by the letter *R* and are printed on our map. Here is their list:

R1. The Neapolis - Scythopolis road (our roads B5-B7);

R2. The road from Ma'aleh Shai to the Jordan Valley. This is a built

road running east-west; crossing a moderate plateau north of Wadi Feiran. Its remains have been well preserved near Kh. Suwede (no. 239). The road connects Wadi Fau with B10 – the Jordan Valley road. It may have originated near Kh. Samrah (no. 129);

R3. From Jabaris to Wadi esh-Shaqq. Its remains were found in Kh. Jabaris itself and near Kh. ej-Jadab (no. 72). This seems to be a local road, connecting Jabaris with the Jordan Valley;

R4. From B8 (The Taiyasir – Meholah road) and B7 (Neapolis – Scythopolis) to Kh. Yarzah



7. A built road, presumably Roman (R7), leading to the Buqei'ah along the eastern edges of Tubas Valley.

(no. 106). Its remains are found northwest of the latter;

R5. From Kh. Yarzah southwards to the Buqei'ah. Its remains are beside Wadi Abu 'Ali (grid 190/192) and Ras el-Ahmar (no. 126);

R6. This approaches Jabaris from the southeast, probably beginning near Hamam el-Malih. Its remains are seen near er-Raqqeh (no. 70);

R7. Coming from Taiyasir (begins at R1) and heading southeast, along the western edges of Ras Jadir. This road passes east of Kh. 'Eynun and descends to the Buqei'ah in a difficult pass. In most of its sections it is well preserved;

R8. South of Beqa'ot, between Umm Rkab and EP 170. It seems to have connected Wadi Far'ah and Kh. Humsah (no. 143). Near EP 141 it splits;

R9. Ascends from Kh. Bet Farr to Tel el-Fukhar (no. 176). This road presumably continued eastwards through the *naqb* near Tel Za'anuni (no. 183) and to the valley of Shechem;

R10. From the Buqei'ah to Kh. Yusef and Kh. es-Samrah. Its remains, found near grid 1948/1870, may have connected roads nos. R5 and R2;

R11. Agricultural roads in the fields of Ras Mubarah, 'Iraq el-Hib and Ras el-Badd;

R12. As above, found in the 'Iraq el-Hamam mass.

In addition, the following roads have been positively defined, though their physical remains have not been found:

a. From Shechem to Kh. Farweh (no. 156), and then to the Jordan Valley. This is our B3, whose remains were blurred due to intensive agricultural cultivation;

b. From Beth Shean to Jericho (our B10). Its remains were detected in the southern Beth Shean Valley. According to the milestones found, it was built during Hadrian's reign (Avi Yonah 1966, 91, and Zertal 1992c; 2005, 28-30).

B. Other Ancient Roads and Paths:

East Manasseh was traversed by four national roads, which connected the Shechem Syncline to the Jordan Valley and/or pass along the syncline's edges.

The topography dictated the road lines until the modern era. It seems that most ancient roads, including the Roman ones, ran along the same lines.

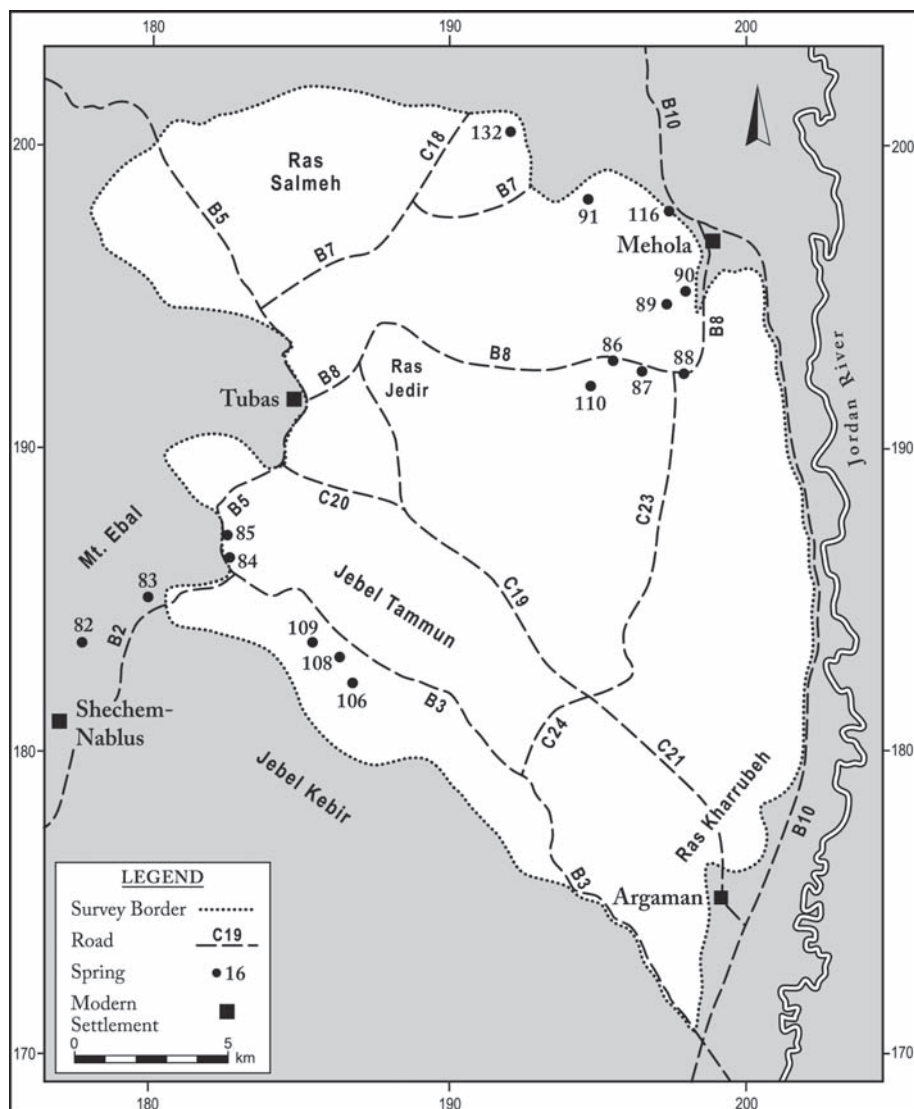
Of the national roads, the following pass through East Manasseh:

1. B2-B3. The Wadi Far'ah road is divided into two parts: B2 – from Shechem to Tel el-Far'ah and B3 – from the latter to Jiftlik. B2 begins in Tel Ballatah (ancient Shechem) and descends along Wadi Abrad to 'Ain

Beidan. Along the road milestones were found, indicating the distances in the Neapolis – Scythopolis road (Thomsen 1917, milestones no. 248-255; Abel 1967, II, 226). The road which succeeded the Roman period might have passed in

a higher level, as indicated by the location of Kh. 'Uqud (site 282 in vol. I, Zertal 2004), where a possible inn stood during the Middle Ages.

At 'Ain Beidan the road splits into two short roads. The first headed



8. The main roads of East Manasseh. B means national road while C – local road. The springs are numbered according to the list in this volume of the survey.

east to Wadi Far'ah and the second north to Tel el-Far'ah. B3 is the Wadi Far'ah road. Today it passes north of the wadi, but according to the SWP map the road passed southwest of the wadi in its lower section, just to cross it near Kh. Basaliyeh (no. 194). It is hard to guess precisely where the road passed in ancient times.

B3 is identified with the "road where the sun goes down" in Deuteronomy 11:30 (*cf.* discussion in Zertal 1988, 316; 1991).

2. B5. From Tel el-Far'ah to the Qabatiyeh junction. This is the eastern parallel to B4 of the first volume. It begins at Tel el-Far'ah and heads north toward Tubas. From there, and through Kh. Fuqahah (no. 47), Kfer and Zebabdeh (nos. 15, 6), it reaches the Qabatiyeh junction. The modern road along the same line crosses the narrow pass beside Kh. 'Anahum, a pass which tends to be flooded in rainy years. Ancient B5 apparently climbed here toward the east to connect to B6, which sets out to the Jezreel Valley.

3. B7, from Kh. Salhab to Beth Shean. It leaves B5 between Fuqahah and 'Aqaba (no. 22). From there, heading northeast, it comes to the *Bezeq* pass, with the two Kh. Ibziq sites (nos. 42, 44) located beside the road. From there, it crossed Sahl Ibziq and turned toward Wadi Khashneh and the Beth Shean Valley. The Roman road here passed directly from Tubas to Taiyasir.

From there to the *Bezeq* pass it is marked C18. Upon reaching Beth Shean it passed near Kh. Qa'un (site no. 3 in the fourth volume of the survey, Zertal 2005). Along B7 several milestones were found; the road was already recorded by the SWP.

4. B8 is the road along Wadi Malih. It leaves B5 at Tubas and heads east toward upper Wadi Malih, called here Wadi el-Hari or, by the SWP, Wadi Sufi el-Khureibat. Then it connects to Wadi Malih near Burj el-Malih. Later on, it leaves the wadi and goes to the fortress at Kh. Umm Ghazal (no. 82) and then to the Jordan Valley.

Of the local roads, the following were added here:

C10 – Meithalun – Kfer: It crosses the Sanur Valley and connects to B5 at Kfer;

C18 – Taiyasir – Kh. Ibziq connects B5 and B7. Here, the remains of the Roman road are clearly visible;

C19 – Taiyasir – the Buqei'ah junction: It leaves Taiyasir along the eastern edges of Tubas Valley. Then it descends to the Buqei'ah and connects to

C20 – Tammun junction – Beqa'ot junction: This road leaves B5 at grid 1845/1905 and moves eastwards to the Buqei'ah. Most of the road is now asphalted, and passes near Kh. 'Atuf (no. 131) to connect to C23 near Kh. Sefireh

(no. 132).

C21 – Buqei'ah junction – Argaman: It crosses the Buqei'ah parallel to C20. After connecting to C23 beside Kh. Umm Qatan it reaches Umm Haraz and descends to the Jordan Valley. This descent is either through Naqb el-Matar (no. 222) or Naqb Umm Khubeza. In both cases it must have reached the valley west of Argaman and, from there, to B10.

C23 – Wadi Far'ah – Tel el-Hilu: This road passes along the modern "Alon road". Then it leaves B3 beside Tel Abu Rumh (no. 190) and ascends to the Buqei'ah to Kh. Muqeysimah. Then it continues northeast to Umm Qatan (no. 133), Kh. Samrah and Wadi Fau. In the PEF map it is indicated "Ancient Road".

C24 – A short branch connecting the eastern Buqei'ah beside Umm Hajar (no. 212), through the Mu-taqallabat valley to Wadi Far'ah.

C. The Landscape Units

(Fig. 9)

In accordance with the principles discussed in the first volume (Zertal 2004, 13-14) East Manasseh was divided into ten landscape units. This division is based on the geomorphology and the history of the area. The order of the units, like order of sites, is from north to south and from west to east.

The landscape units will be dis-

cussed according to two characteristics: their natural data and the settlement type. The first characteristic will be mentioned briefly. In the second section a settlement analysis by periods is presented. In addition, a discussion by periods is found in the recent chapter, section F. We recommend reading this section as a complementary part.

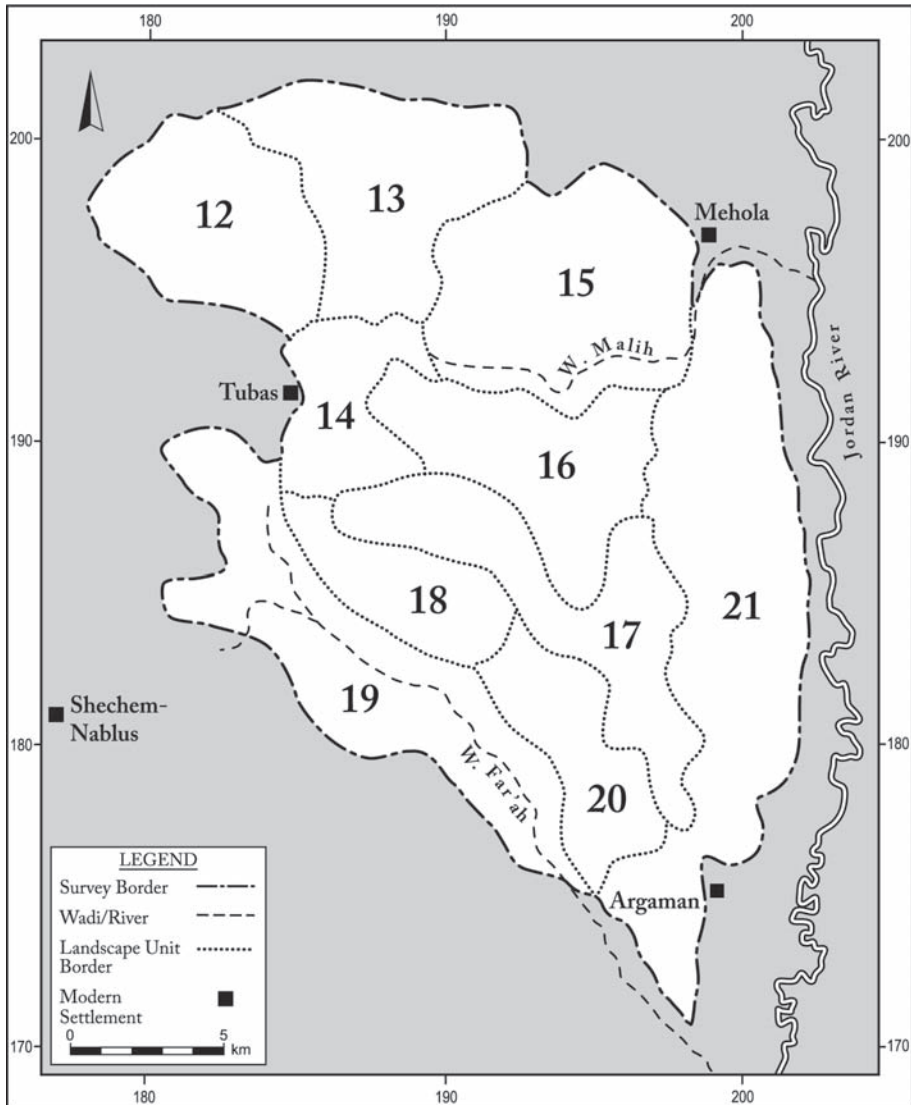
The numbering of the landscape units continues from the first volume. In order to find one's way around, the reader may refer for aid to the borders of the landscape units in figure 9 as well as the attached map of the survey.

12. Zebabdeh Valley

The Zebabdeh Valley covers about 40 sq. km. The valley is, geographically, a highly heterogeneous geographical unit, difficult to clearly determine its natural boundaries. In the north the unit's boundary passes over the Dhahret Tawileh Range, and then to the north of Kh. 'Anahum. From there it turns west to the rocky hills and passes over the upper edge of Sanur Valley, so that the small valley of Sahl Fuheis north of Sir is also included. The southern border passes over the northern edge of Ras el-Aqra' to the village of 'Aqabah, and from there continues along road B5 until Israel Grid 1845/1933. From there it turns east and connects with Landscape Unit no. 13 (Ras es-Salmeh) west of Taiyasir.

Zebabdeh Valley is bordered in the east by the ranges of Ras es-Salmeh and Ras Mubarah (the Far'ah Anticline) and in the west by the eastern edge of Sanur Valley. Its northern part is an easily cultivatable plain, while in the south Wadi Suwed and its tributaries create an

area of low hills. In the western part of the valley is a range (the Zebabdeh Hills), with a small valley (Sahl Fuheis) to its west. The central part of Zebabdeh Valley is covered with deep alluvial soil, which makes it a fertile agricultural region where mostly field crops are grown. In the



9. Map of the landscape units of East Manasseh with their numbers.